

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTMPE405A - NUMERICAL METHODS IN ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
1. Solution in Numerical method are obtained in _____ value
 - a. Exact.
 - b. Approximate
 - c. Imaginary
 - d. Infinite
 2. The bisection method of finding roots of equations falls under the category of _____ method.
 - a. Open
 - b. bracketing
 - c. Random
 - d. graphical
 3. The number 4024.12 contains _____ significant figure
 - a. 5
 - b. 6
 - c. 4
 - d. 7
 4. In an experiment, it is found that the experimental value is very close to actual value, hence the experimental value can be called _____.
 - a. Accurate
 - b. Precise
 - c. Suitable
 - d. Mean
 5. In Gaussian elimination method, original equations are transformed by using _____.
 - a) Column operations
 - b) Row Operations
 - c) Mathematical Operations
 - d) Subset Operation

- 6 If approximate solution of the set of equations, $2x+2y-z=6$, $x+y+2z=8$ and $-x+3y+2z=4$, is given by $x=2.8$, $y=1$ and $z=1.8$. Then, what is the exact solution?
- $x=1, y=3, z=2$
 - $x=2, y=3, z=1$
 - $x=3, y=1, z=2$
 - $x=1, y=2, z=2$
- 7 Numerical techniques more commonly involve _____
- Iterative method
 - Direct method
 - Elimination method
 - Reduction method
- 8 Runge Kutta method is used to solve _____ equation
- Differential
 - Linear
 - Integral
 - Interpolating
- 9 _____ Method used to solve Ordinary differential equation
- Taylor Series
 - bisection
 - Newton Raphson
 - Trapezoidal
- 10 Process of computing intermediate value is called as _____ Method
- Extrapolation
 - Extraction
 - Intermediation
 - Interpolation
- 11 If the data is equally spaced and interpolation is near the beginning of the data then _____ interpolation formula is used
- Newtons Backward difference
 - Newtons Forward difference
 - Lagranges Formula
 - Newtons centre difference
- 12 Newtons Forward interpolation formula is used for _____ intervals
- Open
 - unequal
 - equal
 - closed

Q2. Solve the following.

- Using Newton Raphson method, find a root of the $x^3 + 2x^2 + 10x - 20 = 0$ by taking $x_0 = 0$ 6
- Using Bisection method find the root of equation for $f(x) = x - \cos x$ to four decimal places 6

Q3. Solve the following.

- Solve by using Cramm's Rule $x + 3y + 6z = 23$, $x - y + 4z = 9$, $x - 4y + 2z = 7$ 6
- What is meant by Error, Precision and accuracy? Explain with an example 6

Q4. Solve Any Two of the following.

- A) Evaluate $\int_0^1 \frac{1}{1+x} dx$ the by using Simpsons 1/3 rule 6
- B) Evaluate the $\int_0^{0.5} \frac{x}{\sin x} dx$ using Trapezoidal method 6
- C) A train is moving at the speed of 30 m/s. Suddely the brakes are applied.The speed of the train per second after 't' is given by 6
- | | | | | | | | | | | |
|--------------------|----|----|----|----|----|----|----|----|----|----|
| Time 't' in second | 0 | 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 |
| Speed 'v' in m/s | 30 | 24 | 19 | 16 | 13 | 11 | 10 | 8 | 7 | 5 |
- Estimate the distance moved by the train in 45 seconds

Q5. Solve Any Two of the following.

- A) Using Eurler's Method for $\frac{dy}{dx} = \frac{y-x}{y+x}$ Initial conditions $x_0=0, y_0=1$ find dy/dx at $x=0.1$. Take $h=0.02$ 6
- B) A rod is rotating in a plan. The following table gives the angle (in radians) through which the rod has turned for various values of the time t (seconds). Calculate the angular velocity of the rod at 1 second 6
- | | | | | | | |
|----------------------|---|-----|------|------|------|-------|
| time t (seconds) | 0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 |
| angle (in radians) | 0 | 0.1 | 20.4 | 91.1 | 22.0 | 23.20 |
- C) For value of x ,the respective value of f(x) are as follows, Find f(5) using Newtons finite difference formulax 6
- | | |
|---|------|
| x | f(x) |
| 1 | 14 |
| 2 | 15 |
| 4 | 05 |
| 5 | --- |
| 6 | 09 |

Q6. Solve Any Two of the following.

- A) An object is suspended in a wind tunnel and the force measured for various levels of wind velocity. The results are tabulated below. 6
- | | | | | | | | | |
|--------|----|----|-----|-----|-----|------|-----|------|
| v, m/s | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 |
| F, N | 25 | 70 | 380 | 550 | 610 | 1220 | 830 | 1450 |
- Use least-squares regression to fit this data with (a) a straight line
- B) A fruit seller selling different size mangoes. He sell a small size magno for Rs 3/-, 6 medium size mango for Rs 5/-, and large size mango for Rs 7/-. The seller sells total 15 mangos and has made Rs 77/-. He has sold 2 more medium size mangos than small mangos. How many of each size did he sell.
- C) Find $y(0.2)$ by using Runge-Kutta method taking $h=0.1$ and $y(0)=1$ from the system 6 of equation, $dy/dx = x + y^2$

*** End ***